

## SOCORRO 20 N, NM, USA

WMO: 746340

Lat: 34.3557N

Lon: 106.8859W

Elev: 1477

StdP: 84.79

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 03048

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |             |            |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|-------------|------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |             | 1%         |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-9.9 | (c)<br>-7.8 | (d)<br>-22.0                  | (e)<br>0.6 | (f)<br>10.1 | (g)<br>-19.1 | (h)<br>0.8 | (i)<br>11.2 | (j)<br>7.8            | (k)<br>10.9 | (l)<br>6.7 | (m)<br>9.5 | (n)<br>0.7            | (o)<br>N/A | (p)<br>0.427 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.5                   | 36.7            | 15.0 | 35.6 | 15.1 | 34.3 | 15.0 | 18.7                | 27.0 | 18.2 | 27.0 | 17.8 | 27.0 | 2.9                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.9                            | 14.4 | 20.5 | 16.0 | 13.7 | 20.6 | 15.1 | 12.9 | 21.0 | 59.5          | 26.7 | 57.8 | 26.9 | 56.2 | 26.6 | 21.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 7.0               | 5.9  | 5.2 | -14.2                      | 38.8 | 4.0                | 1.0 | -17.0                                              | 39.5 | -19.4      | 40.0 | -21.6      | 40.6 | -24.5      | 41.3 |  |  |
|                   |      |     | -16.2                      | 19.8 | 4.7                | 0.8 | -19.6                                              | 20.4 | -22.3      | 20.9 | -25.0      | 21.3 | -28.4      | 21.9 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 14.9   | 2.7   | 6.0   | 10.9  | 14.8  | 19.5  | 25.5  | 26.6  | 25.1  | 21.6  | 15.0  | 7.8   | 2.8  |
|                                                                               | DBStd       | 9.19   | 3.84  | 4.28  | 3.99  | 3.84  | 3.87  | 2.56  | 2.24  | 2.13  | 2.88  | 3.78  | 4.05  | 4.26 |
|                                                                               | HDD10.0     | 714    | 227   | 122   | 38    | 8     | 1     | 0     | 0     | 0     | 0     | 6     | 88    | 224  |
|                                                                               | HDD18.3     | 2124   | 484   | 346   | 232   | 114   | 32    | 0     | 0     | 0     | 7     | 114   | 314   | 481  |
|                                                                               | CDD10.0     | 2510   | 1     | 10    | 66    | 153   | 297   | 466   | 516   | 469   | 348   | 161   | 23    | 2    |
|                                                                               | CDD18.3     | 879    | 0     | 0     | 1     | 9     | 69    | 216   | 257   | 211   | 106   | 11    | 0     | 0    |
|                                                                               | CDH23.3     | 11179  | 0     | 3     | 52    | 245   | 1098  | 3052  | 3059  | 2278  | 1177  | 213   | 2     | 0    |
|                                                                               | CDH26.7     | 5415   | 0     | 0     | 3     | 39    | 429   | 1695  | 1645  | 1108  | 458   | 37    | 0     | 0    |
| Wind                                                                          | WSAvg       | 2.1    | 1.7   | 2.1   | 2.5   | 2.9   | 2.6   | 2.4   | 2.1   | 1.9   | 1.9   | 1.9   | 1.8   | 1.7  |
| Precipitation                                                                 | PrecAvg     | 254    | 11    | 8     | 15    | 11    | 15    | 16    | 47    | 44    | 34    | 24    | 11    | 16   |
|                                                                               | PrecMax     | 339    | 88    | 41    | 58    | 62    | 69    | 67    | 101   | 87    | 89    | 61    | 35    | 66   |
|                                                                               | PrecMin     | 111    | 0     | 0     | 0     | 0     | 0     | 2     | 8     | 18    | 1     | 1     | 0     | 0    |
|                                                                               | PrecStd     | 61     | 19    | 9     | 15    | 17    | 17    | 15    | 24    | 23    | 22    | 18    | 11    | 16   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.9  | 23.3  | 26.9  | 29.7  | 34.6  | 38.5  | 38.0  | 36.6  | 34.1  | 29.8  | 23.5  | 18.7 |
|                                                                               |             | MCWB   | 6.3   | 8.5   | 9.7   | 10.9  | 12.5  | 14.5  | 15.8  | 15.4  | 15.2  | 12.6  | 9.1   | 7.2  |
|                                                                               | 2%          | DB     | 15.1  | 20.2  | 24.9  | 27.9  | 32.7  | 37.0  | 36.8  | 35.2  | 32.7  | 27.6  | 21.5  | 16.1 |
|                                                                               |             | MCWB   | 5.4   | 7.3   | 8.8   | 10.1  | 12.1  | 14.3  | 15.9  | 16.1  | 15.0  | 12.3  | 8.7   | 6.0  |
|                                                                               | 5%          | DB     | 13.2  | 17.9  | 22.8  | 26.2  | 31.0  | 35.7  | 35.6  | 33.9  | 31.2  | 25.8  | 19.2  | 13.6 |
|                                                                               |             | MCWB   | 4.5   | 6.2   | 8.0   | 9.6   | 11.5  | 13.9  | 16.2  | 16.2  | 14.8  | 11.7  | 7.9   | 5.1  |
|                                                                               | 10%         | DB     | 11.1  | 15.4  | 20.7  | 24.6  | 29.2  | 34.3  | 34.1  | 32.6  | 29.6  | 23.9  | 16.8  | 11.3 |
|                                                                               |             | MCWB   | 3.4   | 5.3   | 7.1   | 9.0   | 10.9  | 13.7  | 16.1  | 16.3  | 14.5  | 11.0  | 7.1   | 4.2  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.0   | 9.4   | 10.6  | 12.4  | 14.7  | 17.8  | 19.6  | 19.2  | 18.3  | 17.2  | 11.2  | 8.5  |
|                                                                               |             | MCDB   | 16.3  | 21.3  | 24.1  | 24.8  | 28.1  | 29.5  | 26.3  | 27.5  | 25.1  | 22.9  | 17.2  | 15.1 |
|                                                                               | 2%          | WB     | 6.2   | 8.1   | 9.5   | 11.3  | 13.3  | 16.9  | 18.9  | 18.6  | 17.6  | 15.2  | 10.0  | 7.2  |
|                                                                               |             | MCDB   | 13.8  | 19.3  | 23.0  | 24.1  | 26.9  | 30.1  | 26.7  | 27.5  | 24.9  | 22.0  | 16.7  | 14.4 |
|                                                                               | 5%          | WB     | 5.1   | 7.1   | 8.6   | 10.5  | 12.4  | 16.2  | 18.4  | 18.1  | 17.0  | 13.7  | 8.9   | 6.0  |
|                                                                               |             | MCDB   | 11.8  | 17.4  | 21.7  | 23.6  | 26.4  | 29.9  | 27.0  | 27.2  | 24.7  | 21.8  | 16.8  | 12.5 |
|                                                                               | 10%         | WB     | 4.0   | 6.0   | 7.8   | 9.8   | 11.6  | 15.5  | 17.9  | 17.7  | 16.5  | 12.4  | 7.7   | 4.9  |
|                                                                               |             | MCDB   | 10.3  | 15.4  | 20.4  | 23.1  | 25.9  | 29.5  | 27.2  | 26.9  | 24.6  | 20.9  | 16.2  | 11.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.9  | 16.1  | 16.6  | 16.2  | 16.1  | 17.0  | 14.5  | 14.2  | 14.7  | 15.6  | 15.9  | 14.1 |
|                                                                               |             | MCDBR  | 18.1  | 20.1  | 19.8  | 18.2  | 18.2  | 18.3  | 16.1  | 16.0  | 16.7  | 17.7  | 18.7  | 17.3 |
|                                                                               |             | MCWBR  | 10.6  | 10.0  | 9.4   | 7.5   | 7.1   | 6.4   | 4.0   | 4.2   | 5.3   | 7.5   | 9.1   | 9.7  |
|                                                                               | 5% WB       | MCDBR  | 16.1  | 18.9  | 18.6  | 16.9  | 16.8  | 16.0  | 13.3  | 13.5  | 12.4  | 13.9  | 15.1  | 15.0 |
|                                                                               |             | MCWBR  | 9.2   | 9.4   | 8.7   | 6.8   | 6.4   | 5.1   | 3.4   | 3.5   | 4.0   | 6.0   | 7.9   | 8.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.241  | 0.248 | 0.270 | 0.291 | 0.308 | 0.329 | 0.370 | 0.346 | 0.312 | 0.278 | 0.254 | 0.244 |      |
|                                                                               | taud        | 2.638  | 2.619 | 2.540 | 2.462 | 2.464 | 2.399 | 2.395 | 2.427 | 2.517 | 2.585 | 2.599 | 2.624 |      |
|                                                                               | Ebn at Noon | 999    | 1020  | 1013  | 996   | 974   | 948   | 908   | 927   | 951   | 969   | 970   | 974   |      |
|                                                                               | Edh at Noon | 72     | 82    | 97    | 110   | 112   | 119   | 119   | 113   | 98    | 84    | 74    | 69    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 3.35   | 4.34  | 5.72  | 6.98  | 7.76  | 7.93  | 7.05  | 6.50  | 5.81  | 4.75  | 3.67  | 2.98  |      |
|                                                                               | RadStd      | 0.27   | 0.35  | 0.34  | 0.34  | 0.39  | 0.41  | 0.36  | 0.22  | 0.24  | 0.37  | 0.21  | 0.17  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.67 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +155    | +87     |
| (47) Regional (12 neighbors) |  | +0.30 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +92     | +52     |

Nomenclature: See separate page